

Beta Blockers

- Pharmacology study group
- Cardiovascular drugs, week 4

Mechanism of action

- Block beta-adrenergic receptors, competing with adrenaline and noradrenaline
- Beta-1 blockade in the heart lowers heart rate and contractility
- Reduce renin release from the kidney, which lowers blood pressure
- Slow conduction through the AV node

Classification

- Non-selective (beta-1 and beta-2): propranolol, timolol
- Cardioselective (beta-1): atenolol, metoprolol, bisoprolol
- With alpha-blocking activity: carvedilol, labetalol
- Cardioselectivity is lost at high doses

Clinical uses

- Angina, and after myocardial infarction to reduce mortality
- Stable heart failure: bisoprolol, carvedilol, metoprolol succinate
- Rate control in atrial fibrillation
- Propranolol for migraine prophylaxis, essential tremor and thyrotoxicosis
- Timolol eye drops for glaucoma

Adverse effects and contraindications

- Bradycardia, hypotension, fatigue and cold extremities
- Bronchospasm: non-selective agents are contraindicated in asthma
- Can mask the warning signs of hypoglycaemia in diabetics
- Never stop abruptly: risk of rebound tachycardia and angina
- Avoid in second or third degree heart block